

Лабораторная работа №2

Управление пользователями и группами

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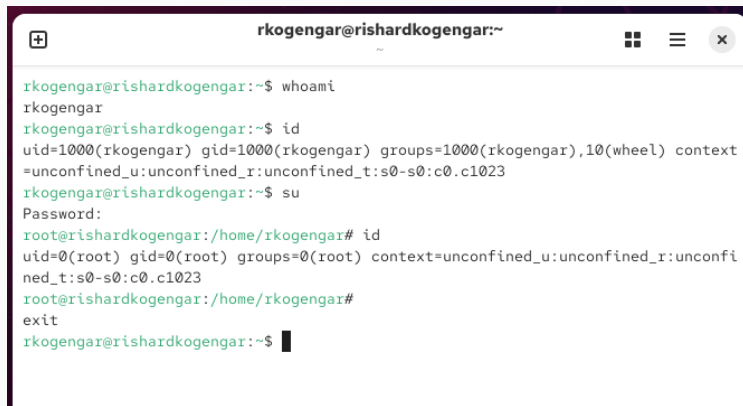
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Цель работы

Получить представление о работе с учётными записями пользователей и группами пользователей в операционной системе типа Linux.

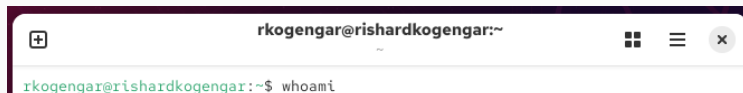
Ход выполнения работы

Проверка пользователя и переход на root



```
rkogengar@rishardkogengar:~  
  
rkogengar@rishardkogengar:~$ whoami  
rkogengar  
rkogengar@rishardkogengar:~$ id  
uid=1000(rkogengar) gid=1000(rkogengar) groups=1000(rkogengar),10(wheel) context=  
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
rkogengar@rishardkogengar:~$ su  
Password:  
root@rishardkogengar:/home/rkogengar# id  
uid=0(root) gid=0(root) groups=0(root) context=unconfined_u:unconfined_r:unconfi  
ned_t:s0-s0:c0.c1023  
root@rishardkogengar:/home/rkogengar#  
exit  
rkogengar@rishardkogengar:~$
```

Рис. 1: Проверка пользователя rkogengar



```
rkogengar@rishardkogengar:~  
  
rkogengar@rishardkogengar:~$ whoami
```

Настройка sudoers

```
rkogengar@rishardkogengar:~ – sudo -i visudo

## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##
##      user      MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)        ALL

## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
# %sys ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING, PROCESSES, LOC
ATE, DRIVERS

## Allows people in group wheel to run all commands
%wheel  ALL=(ALL)        ALL

## Same thing without a password
# %wheel          ALL=(ALL)        NOPASSWD: ALL
```

105,0-1 89%

```
rkogengar@rishardkogengar:~$ sudo -i useradd -G wheel alice
rkogengar@rishardkogengar:~$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),10(wheel)
rkogengar@rishardkogengar:~$ sudo -i passwd alice
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
rkogengar@rishardkogengar:~$ su alice
Password:
alice@rishardkogengar:/home/rkogengar$ sudo useradd bob
```

Рис. 4: Создание пользователя alice

Создание пользователей

```
rkogengar@rishardkogengar:~$ su alice
Password:
alice@rishardkogengar:/home/rkogengar$ sudo useradd bob

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

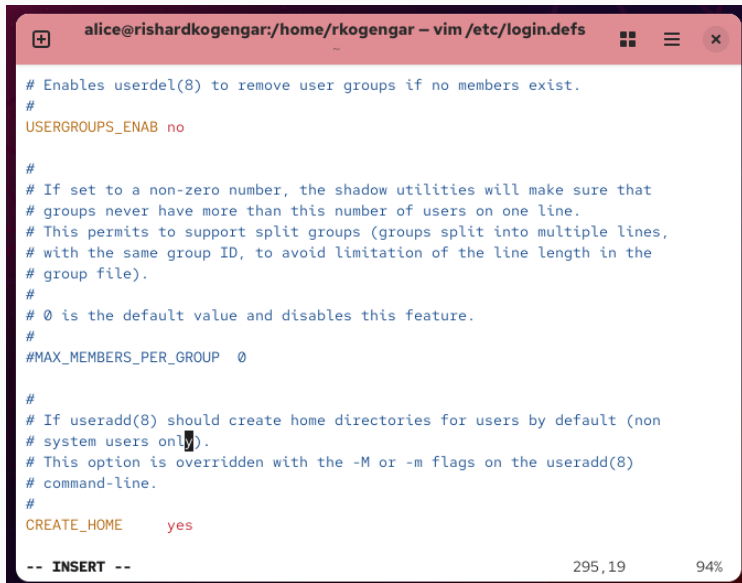
    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for alice:
alice@rishardkogengar:/home/rkogengar$ sudo passwd bob
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
alice@rishardkogengar:/home/rkogengar$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)
alice@rishardkogengar:/home/rkogengar$
```

Рис. 5: Создание пользователя bob

Настройка login.defs



```
alice@rishardkogengar:/home/rkogengar - vim /etc/login.defs

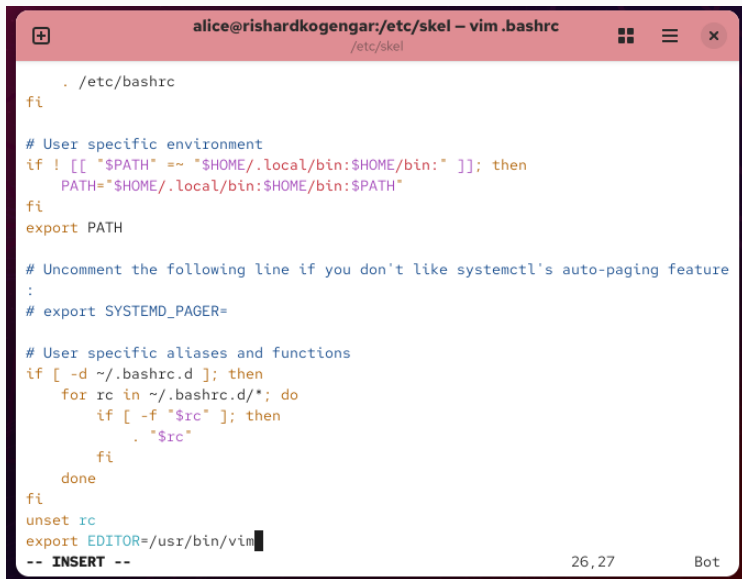
# Enables userdel(8) to remove user groups if no members exist.
#
USERGROUPS_ENAB no

#
# If set to a non-zero number, the shadow utilities will make sure that
# groups never have more than this number of users on one line.
# This permits to support split groups (groups split into multiple lines,
# with the same group ID, to avoid limitation of the line length in the
# group file).
#
# 0 is the default value and disables this feature.
#
#MAX_MEMBERS_PER_GROUP 0

#
# If useradd(8) should create home directories for users by default (non
# system users only).
# This option is overridden with the -M or -m flags on the useradd(8)
# command-line.
#
CREATE_HOME yes

-- INSERT --
```

Изменение .bashrc



The screenshot shows a terminal window with the title bar "alice@rishardkogengar:/etc/skel – vim .bashrc". The window displays the contents of the .bashrc file. The code is as follows:

```
. /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]; then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature
:
# export SYSTEMD_PAGER=

# User specific aliases and functions
if [ -d ~/.bashrc.d ]; then
    for rc in ~/.bashrc.d/*; do
        if [ -f "$rc" ]; then
            . "$rc"
        fi
    done
fi
unset rc
export EDITOR=/usr/bin/vim
-- INSERT --
```

At the bottom right of the terminal window, the text "26,27" and "Bot" are visible.

Создание пользователя carol

```
alice@rishardkogengar:/etc/skel$ sudo -i useradd carol
[sudo] password for alice:
alice@rishardkogengar:/etc/skel$ sudo passwd carol
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
alice@rishardkogengar:/etc/skel$ su carol
Password:
carol@rishardkogengar:/etc/skel$ id
uid=1003(carol) gid=100(users) groups=100(users) context=unconfined_u:unconfined
_r:unconfined_t:s0-s0:c0.c1023
carol@rishardkogengar:/etc/skel$ cd
carol@rishardkogengar:~$ ls -
ls: cannot access '-': No such file or directory
carol@rishardkogengar:~$ ls -Al
total 12
-rw-r--r--. 1 carol users 18 Oct 29 2024 .bash_logout
-rw-r--r--. 1 carol users 144 Oct 29 2024 .bash_profile
-rw-r--r--. 1 carol users 549 Sep 8 19:27 .bashrc
drwxr-xr-x. 2 carol users 6 Sep 8 19:25 Documents
drwxr-xr-x. 4 carol users 39 Sep 4 18:16 .mozilla
drwxr-xr-x. 2 carol users 6 Sep 8 19:25 Pictures
carol@rishardkogengar:~$
```

Рис. 8: Создание пользователя carol

Настройка срока действия пароля

```
-----@-----gengar:~$ su alice
carol@rishardkogengar:~$ su alice
Password:
alice@rishardkogengar:/home/carol$ sudo cat /etc/shadow | grep carol
carol:$y$j9T$PtHQi2tuJPLzLkYkqjTIT0$lE0tmXA/IKe4L/uqJt3zY/0FEBcLy7zH8TUdK5kqqM0:
20339:0:99999:7:::
alice@rishardkogengar:/home/carol$ sudo passwd -n 30 -w 3 -x 90 carol
passwd: password changed.
alice@rishardkogengar:/home/carol$ sudo cat /etc/shadow | grep carol
carol:$y$j9T$PtHQi2tuJPLzLkYkqjTIT0$lE0tmXA/IKe4L/uqJt3zY/0FEBcLy7zH8TUdK5kqqM0:
20339:30:90:3:::
alice@rishardkogengar:/home/carol$
```

Рис. 9: Настройка срока действия пароля

```
alice@rishardkogengar:/home/carol$  
alice@rishardkogengar:/home/carol$ sudo grep alice /etc/passwd /etc/shadow /etc/  
group  
/etc/passwd:alice:x:1001:1001::/home/alice:/bin/bash  
/etc/shadow:alice:$y$j9T$tjYFTS3SMjgLv0vPtWQi1$TyGviKRY/BNgkDqkoXZdRX4ImJO.KvHZ  
0HZLnpoWPT7:20339:0:99999:7:::  
/etc/group:wheel:x:10:rkogengar,alice  
/etc/group:alice:x:1001:  
alice@rishardkogengar:/home/carol$ sudo grep carol /etc/passwd /etc/shadow /etc/  
group  
/etc/passwd:carol:x:1003:100::/home/carol:/bin/bash  
/etc/shadow:carol:$y$j9T$PtHQi2tuJPLzLkYkqjTIT0$lE0tmXA/IKe4L/uqJt3zY/0FEBcLy7zH  
8TUdK5kqqM0:20339:30:90:3:::  
alice@rishardkogengar:/home/carol$
```

Рис. 10: Проверка записей о пользователях

```
alice@rishardkogengar:/home/carol$  
alice@rishardkogengar:/home/carol$  
alice@rishardkogengar:/home/carol$ sudo usermod -aG main alice  
alice@rishardkogengar:/home/carol$ sudo usermod -aG main bob  
alice@rishardkogengar:/home/carol$ sudo usermod -aG third carol  
alice@rishardkogengar:/home/carol$ id carol  
uid=1003(carol) gid=100(users) groups=100(users),1004(third)  
alice@rishardkogengar:/home/carol$ id alice  
uid=1001(alice) gid=1001(alice) groups=1001(alice),10(wheel),1003(main)  
alice@rishardkogengar:/home/carol$ id bob  
uid=1002(bob) gid=1002(bob) groups=1002(bob),1003(main)  
alice@rishardkogengar:/home/carol$
```

Рис. 11: Проверка пользователей и групп

Итоги работы

В ходе работы были изучены основы управления пользователями и группами в Linux: создание учётных записей, настройка паролей, редактирование sudoers и работа с группами. Система успешно установлена и проверена в VirtualBox.